Regression Discontinuity

Raffaele Saggio

UBC

October 28, 2026

Introduction

- The **regression discontinuity design** (RD) is one of the most popular quasi-experimental research designs in contemporary applied research
- RD methods can be applied when a researcher has specific information about the rules determining the treatment of interest
- This lecture describes the RD framework and implementation
- See MHE; Imbens and Lemieux (2008), Lee and Lemieux (2010), and Imbens and Wooldridge (2009) for recent overviews of RD

Sharp RD: Basic Setup

- Consider a setting with a treatment $T_i \in \{0, 1\}$, and potential outcomes $Y_i(1)$ and $Y_i(0)$
- Suppose the treatment is a deterministic function of an observed covariate R_i, such that

$$T_i = 1\{R_i > c\}.$$

- R_i is called the **running variable**
- This is a **sharp RD** because the probability of treatment switches from zero to one at the threshold
- Example: Scholarship awarded to students who score above a test score threshold (Thistlethwaite and Campbell 1960)

Sharp RD: Basic Setup

- We get to observe $Y_i(1)$ when $R_i > c$ and $Y_i(0)$ when $R_i \leq c$
- Basic idea of RD design: Compare observations just above and just below the threshold to infer treatment effect
- Intuitively, the treatment may be as good as randomly assigned for individuals in the neighborhood of $R_i = c$, so comparing treated and nontreated reveals a treatment effect

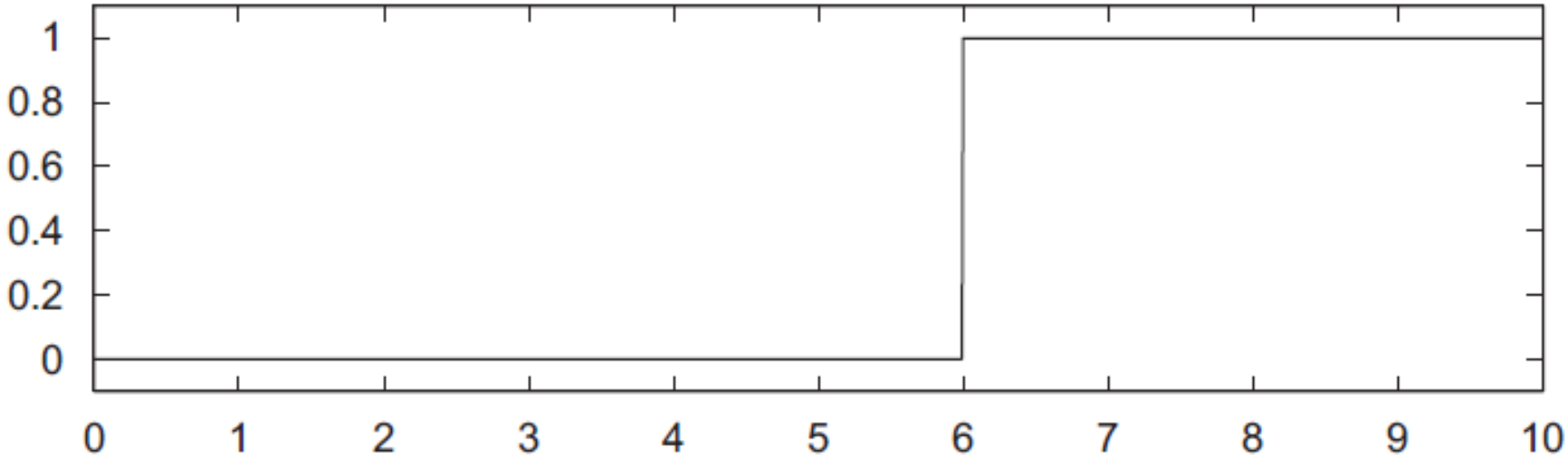


Fig. 1. Assignment probabilities (SRD).

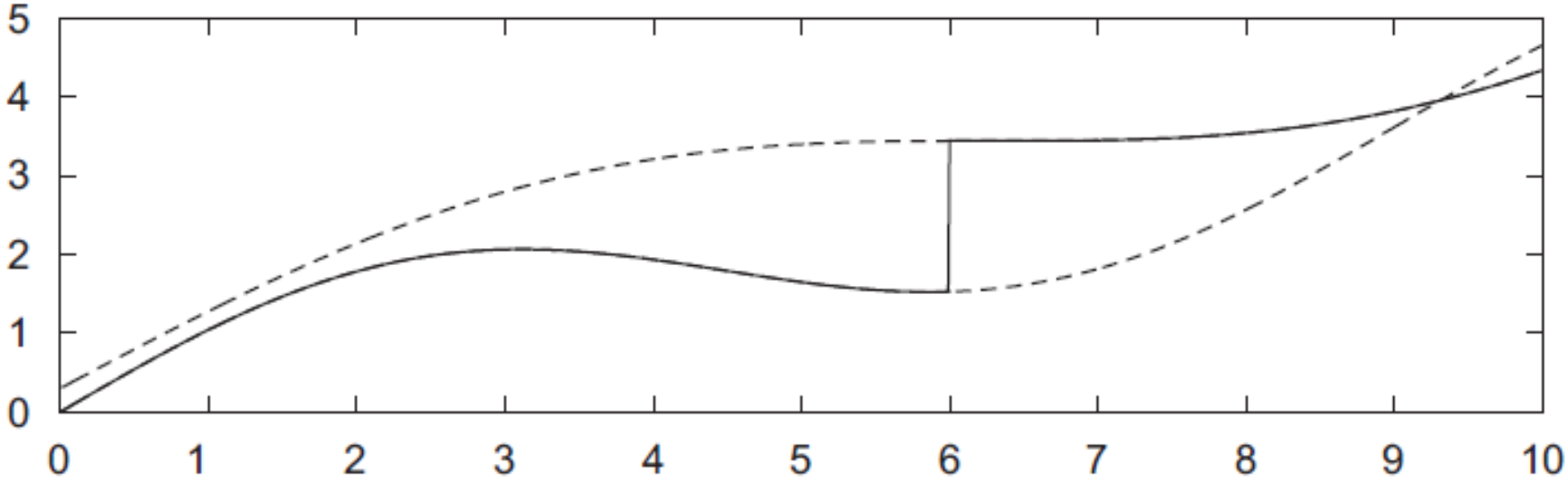


Fig. 2. Potential and observed outcome regression functions.

Sharp RD Identification

- Key RD identifying assumption: potential outcomes are smooth at the threshold
- Formally:

$$\lim_{r \to c^+} E\left[Y_i(t) | R_i = r\right] = \lim_{r \to c^-} E\left[Y_i(t) | R_i = r\right], \ t \in \{0, 1\}$$

- Potential outcome CEFs must be continuous at the threshold
- The population just below must not be discretely different from the population just above
- The only thing that is allowed to change disproportionately at the threshold is the assignment probability of treatment

Sharp RD Identification

- If this assumption holds we have

$$\lim_{r \to c^+} E\left[Y_i | R_i = r\right] - \lim_{r \to c^-} E\left[Y_i | R_i = r\right]$$

$$= \lim_{r \to c^+} E\left[Y_i(1) | R_i = r\right] - \lim_{r \to c^-} E\left[Y_i(0) | R_i = r\right]$$

$$= E\left[Y_i(1) | R_i = c\right] - E\left[Y_i(0) | R_i = c\right]$$

$$= E\left[Y_i(1) - Y_i(0) | R_i = c\right]$$

- When potential outcomes are smooth around the threshold, a comparison of individuals just above and just below yields the average treatment effect for individuals at the threshold

- This identification argument is nonparametric: we didn't have to assume anything about the distribution of $Y_i(t)$ other than continuity of CEFs

Sharp RD Interpretation

- Note that in a sharp RD design we have the CIA condition

$$(Y_i(1), Y_i(0)) \perp T_i | R_i$$

- CIA holds trivially because there is no variation in T_i conditional on R_i
- But there is also no common support – T_i is either always zero or always one conditional on R_i
- RD estimation is a local extrapolation outside the support of the data to predict the mean treated and untreated potential outcomes at $R_i = c$

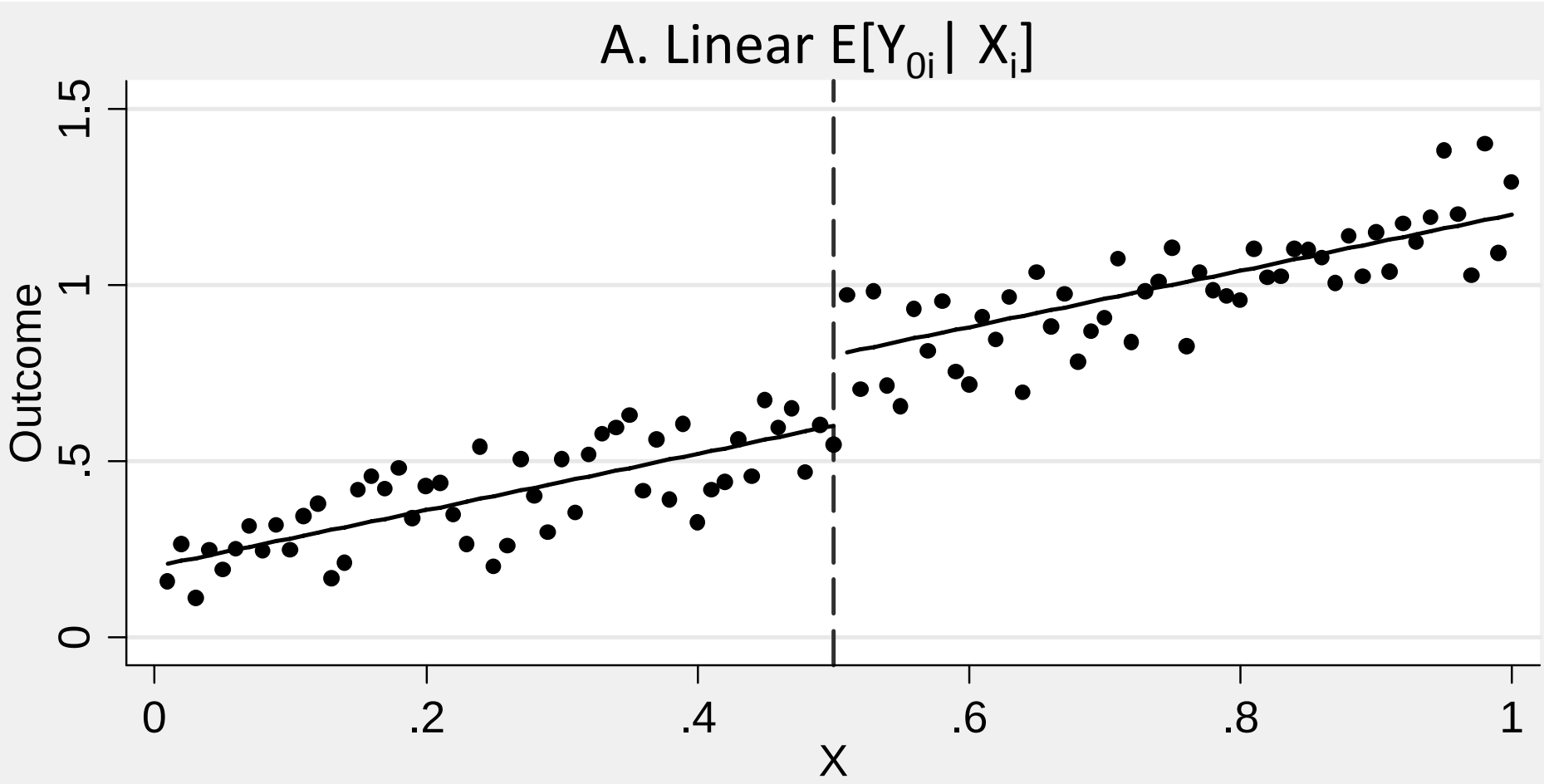
RD Estimation

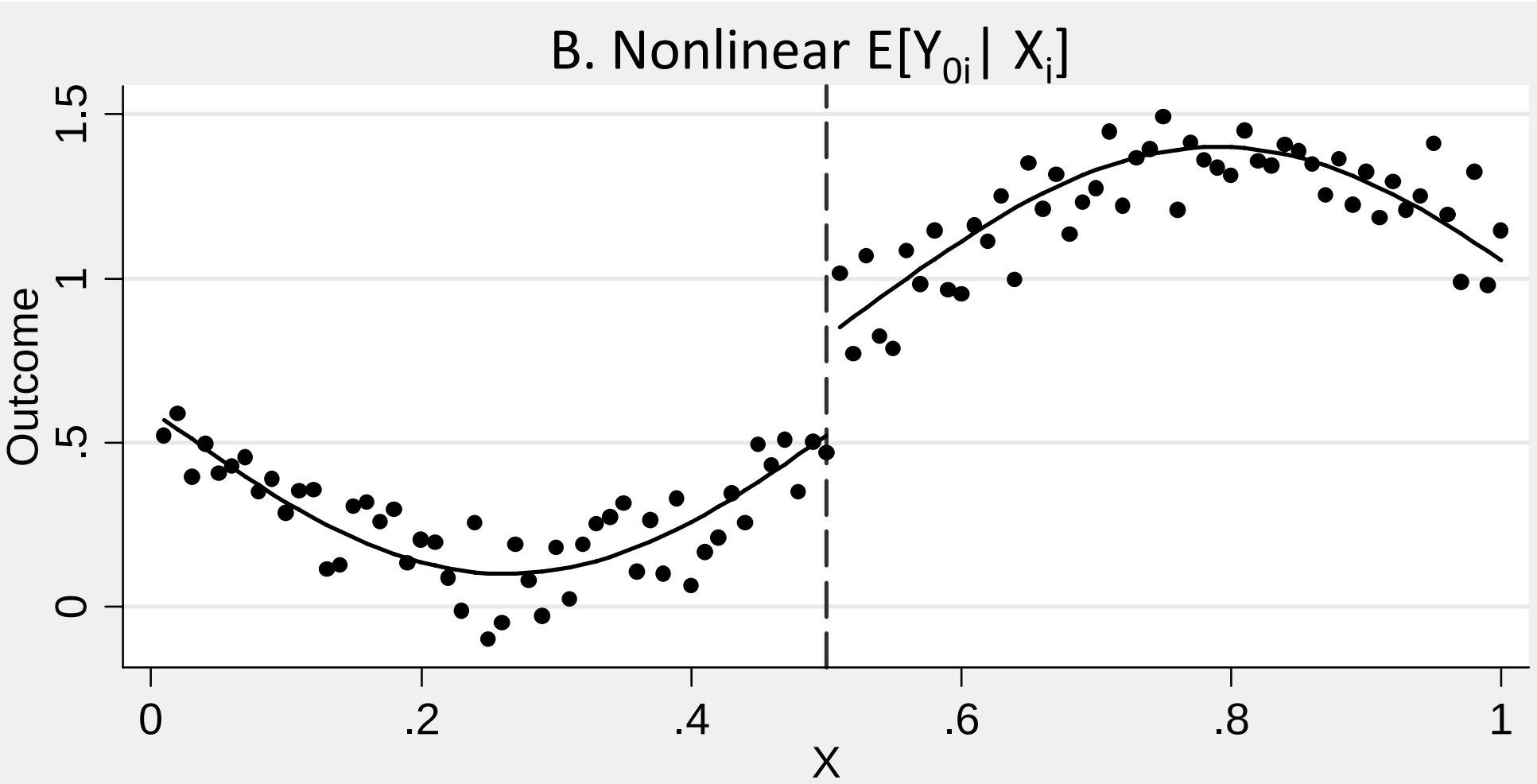
- Implementing a sharp RD requires estimating the right- and left-hand limits

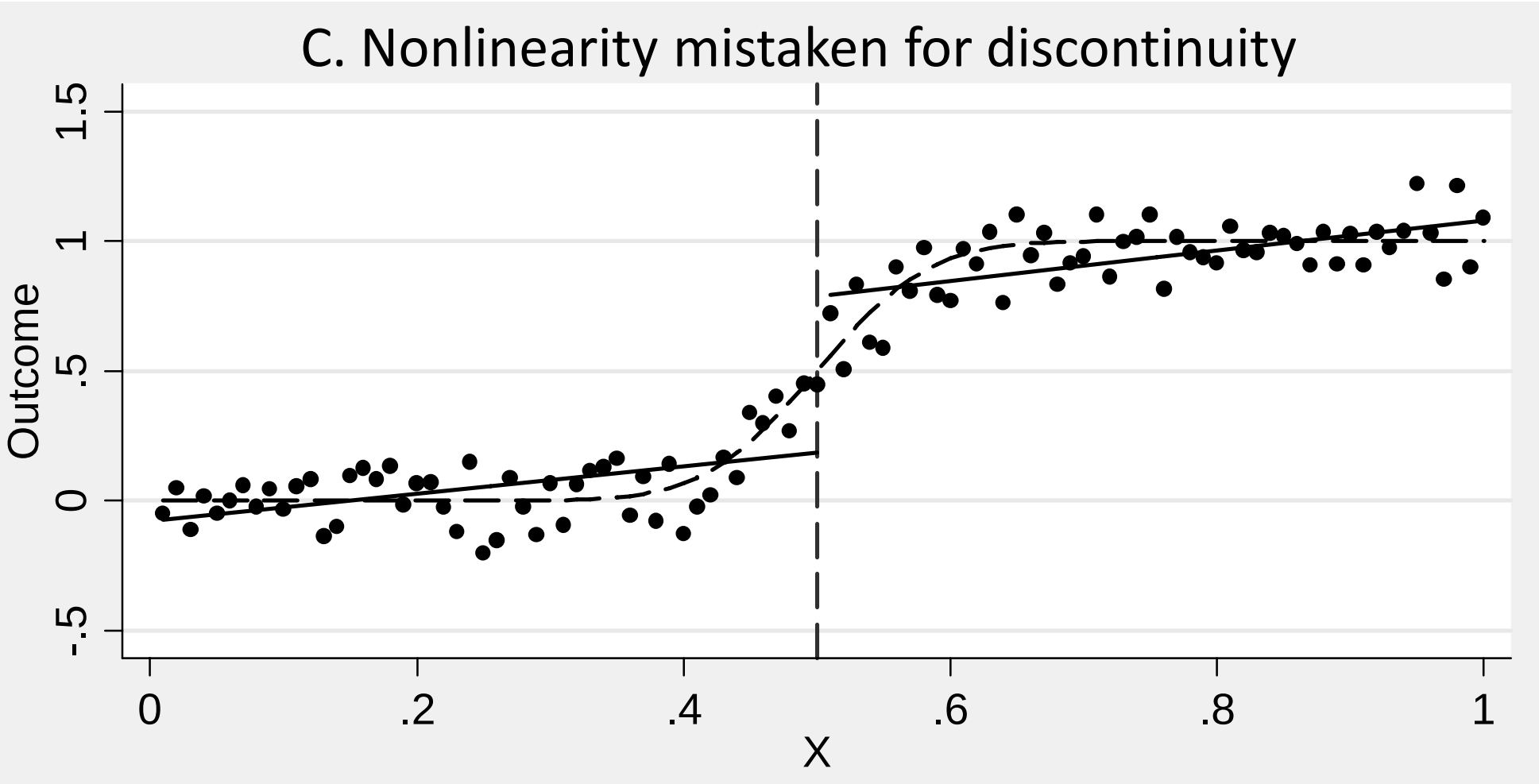
$$\lim_{r \to c^+} E\left[Y_i | R_i = r\right]$$

$$\lim_{r \to c^-} E\left[Y_i | R_i = r\right]$$

- Since extrapolation is central to the RD design, functional form used to approximate $E\left[Y_i | R_i\right]$ really matters
- Not enough to just do OLS and rely on approximation theorems
- Insufficiently flexible specification of the CEF runs the risk of mistaking nonlinearity for treatment effect
- But overly flexible specification reduces precision and runs the risk of overfitting


A. Linear E[Y0i| Xi]
Outcome
0
.5
1
1.5
0
.2
.4
.6
.8
1
X


B. Nonlinear E[Y0i| Xi]
Outcome
0
.5
1
1.5
0
.2
.4
.6
.8
1
X


C. Nonlinearity mistaken for discontinuity
Outcome
-.5
0
.5
1
1.5
0
.2
.4
.6
.8
1
X

RD Estimation

- Two general approaches to estimation:
  - Global parametric
  - Local nonparametric

Global Polynomial

- The global approach uses parametric functions to approximate $E\left[Y_i|R_i\right]$, typically polynomials
- Estimating equation:

$$Y_i = \alpha + \beta 1\left\{R_i > c\right\} + \sum_{k=1}^{K} \gamma_{0k} \left(R_i - c\right)^k + \sum_{k=1}^{K} \gamma_{1k} 1\left\{R_i < c\right\} \left(R_i - c\right)^k + \epsilon_i$$

- This specification uses a K-th order polynomial with coefficients that differ on each side of the threshold
- The parameter β measures the jump at the threshold

Local Linear Regression

- More modern approach: use nonparametric techniques to approximate the left- and right-hand limits
- Local linear regression:

$$\left(\hat{\alpha}_0, \hat{\delta}_0\right) = \arg\min_{\alpha_0, \delta_0} \sum_i 1\left\{R_i \leq c\right\} K\left(\frac{R_i - c}{h}\right) \left(Y_i - \alpha_0 - \delta_0 \left(R_i - c\right)\right)^2$$

- Here $K(\cdot)$ is a symmetric kernel function maximized at 0, and h is a bandwidth that vanishes asymptotically

Local Linear Regression

$$\left(\hat{\alpha}_0, \hat{\delta}_0\right) = \arg\min_{\alpha_0, \delta_0} \sum_i 1\left\{R_i \leq c\right\} K\left(\frac{R_i - c}{h}\right) \left(Y_i - \alpha_0 - \delta_0 \left(R_i - c\right)\right)^2$$

- This is weighted least squares using observations to the left of the threshold, and weighting by proximity to the threshold
- The bandwidth h determines how quickly weight falls off away from the discontinuity

Local Linear Regression

$$\left(\hat{\alpha}_0, \hat{\delta}_0\right) = \arg\min_{\alpha_0, \delta_0} \sum_i 1\left\{R_i \leq c\right\} K\left(\frac{R_i - c}{h}\right) \left(Y_i - \alpha_0 - \delta_0 \left(R_i - c\right)\right)^2$$

- $\hat{\alpha}_0$ is an estimate of $\lim_{r \to c^-} E\left[Y_i | R_i = r\right]$
- Run similar regression for the sample with $R_i > c$ to get $\hat{\alpha}_1$
- Treatment effect estimate is $\hat{\alpha}_1 - \hat{\alpha}_0$

RD Implementation Issues

- Remaining questions:
  - Global or local?
  - How to choose order of polynomial?
  - How to choose bandwidth?
- We will come back to these issues

Fuzzy RD

- In reality treatment is often generated by a discontinuous assignment rule that isn't deterministic
- Suppose that

$$\lim_{r \to c^-} Pr\left[T_i = 1 | R_i = r\right] \neq \lim_{r \to c^+} Pr\left[T_i = 1 | R_i = r\right]$$

- The probability of treatment jumps at $R_i = c$, but not necessarily from zero to one
- This is a **fuzzy RD** scenario because treatment is only partly determined by the threshold
- Example: An income threshold determines eligibility for a government program, but not every eligible household participates

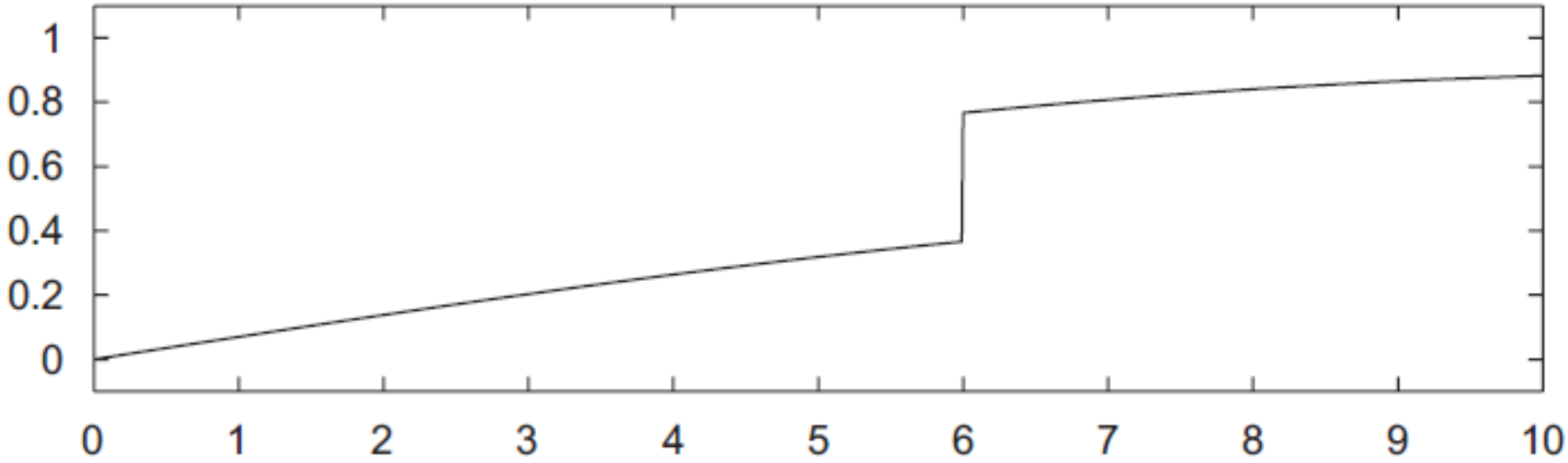


Fig. 3. Assignment probabilities (FRD).

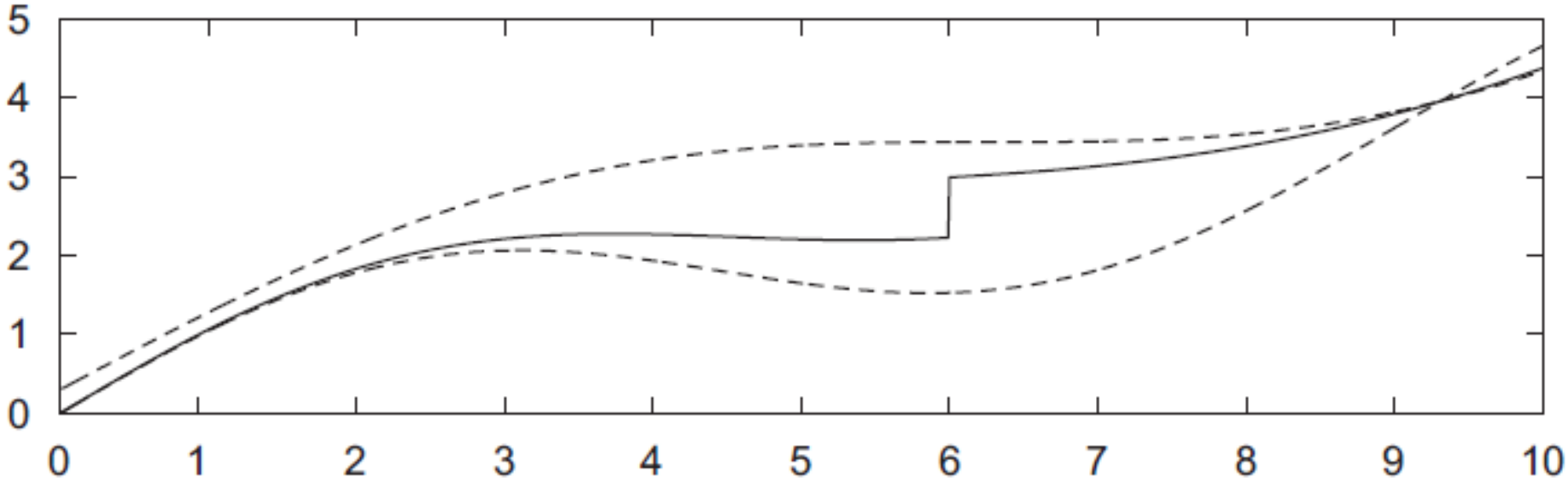


Fig. 4. Potential and observed outcome regression (FRD).

Fuzzy RD Assumptions

- As before, assume the distributions of $Y_i(1)$ and $Y_i(0)$ are smooth around the threshold
- Let $T_i(1)$ and $T_i(0)$ denote potential treatment statuses for individual i if s/he were located above and below the threshold. Assume these are also smooth across the threshold, and

$$T_i(1) \geq T_i(0) \ \forall i$$

- Crossing the threshold weakly increases the probability of treatment for everyone

Fuzzy RD

- Again compare individuals above and below the threshold:

$$\lim_{r \to c^+} E\left[Y_i | R_i = r\right] - \lim_{r \to c^-} E\left[Y_i | R_i = r\right]$$

$$= \lim_{r \to c^+} E\left[Y_i(0) + \left(Y_i(1) - Y_i(0)\right) T_i | R_i = r\right]$$

$$- \lim_{r \to c^-} E\left[Y_i(0) + \left(Y_i(1) - Y_i(0)\right) T_i | R_i = r\right]$$

$$= \lim_{r \to c^+} E\left[\left(Y_i(1) - Y_i(0)\right) T_i(1) | R_i = r\right] - \lim_{r \to c^-} E\left[\left(Y_i(1) - Y_i(0)\right) T_i(0) | R_i = r\right]$$

$$= E\left[\left(Y_i(1) - Y_i(0)\right)\left(T_i(1) - T_i(0)\right) | R_i = c\right]$$

Fuzzy RD

- Using monotonicity we can rewrite this as

$$= E\left[\left(Y_i(1) - Y_i(0)\right)\left(T_i(1) - T_i(0)\right)|R_i = c\right]$$

$$= E\left[Y_i(1) - Y_i(0)|T_i(1) > T_i(0), R_i = c\right] \times Pr\left[T_i(1) > T_i(0)|R_i = c\right]$$

- The compliance probability at the threshold is given by

$$\lim_{r \to c^+} E\left[T_i|R_i = r\right] - \lim_{r \to c^-} E\left[T_i|R_i = r\right] = Pr\left[T_i(1) > T_i(0)|R_i = c\right]$$

Fuzzy RD

- Then by taking a ratio we obtain

$$\frac{\lim_{r \to c^+} E\left[Y_i | R_i = r\right] - \lim_{r \to c^-} E\left[Y_i | R_i = r\right]}{\lim_{r \to c^+} E\left[T_i | R_i = r\right] - \lim_{r \to c^-} E\left[T_i | R_i = r\right]} =$$
$$E\left[Y_i(1) - Y_i(0) \middle| T_i(1) > T_i(0), R_i = c\right]$$

- The ratio of the jump in the outcome CEF to the jump in the treatment probability identifies an average treatment effect for individuals who switch treatment status at the threshold

- Look familiar?

Fuzzy RD is IV

- Fuzzy RD is IV using a threshold indicator $Z_i = 1\{R_i > c\}$ as an instrument for treatment in the neighborhood of the threshold

RD is doubly LATE

- This implies fuzzy RD estimates are local in two senses
- First, they are local to the threshold, $R_i = c$
  - Also applies to strict RD estimates
- Second, they apply only to compliers at the threshold, rather than everyone with $R_i = c$
  - This is the "local" in LATE

Fuzzy RD Implementation

- Can implement fuzzy RD with a global polynomial 2SLS approach:

$$T_i = \lambda + \pi 1\{R_i > c\} + \sum_{k=1}^{K} \theta_{0k}(R_i - c)^k + \sum_{k=1}^{K} \theta_{1k} 1\{R_i > c\}(R_i - c)^k + \eta_i$$

$$Y_i = \alpha + \beta \hat{T}_i + \sum_{k=1}^{K} \gamma_{0k}(R_i - c)^k + \sum_{k=1}^{K} \gamma_{1k} 1\{R_i > c\}(R_i - c)^k + \epsilon_i$$

- Excluded instrument is $1\{R_i > c\}$
- Alternatively, can estimate each of the four limits in the Wald ratio by local linear regressions of Y_i and T_i on R_i

Visual RD

- RD is appealing because the design is relatively simple and transparent
- Key relationships are between the outcome and running variable, and treatment and running variable
- These relationships can be plotted, and one can look visually for evidence of a discontinuity
- In practice people won't believe an RD estimate if the result isn't visible in a plot
  - Automated plotting procedure: "rdbinselect" (Calonico, Cattaneo and Titiunik, 2014)

RD Diagnostics

- Fundamental RD identifying assumption: potential outcome distributions are smooth around the threshold
- $1\{R_i > c\}$ must be as good as randomly assigned in the neighborhood of $R_i = c$
- May be violated if individuals can exactly control the value of R_i and therefore location relative to the threshold
- Example: suppose income is self-reported, and people intentionally misreport to gain program eligibility
- Identifying assumption is untestable, but some common diagnostics can serve as a guide to its plausibility

RD Diagnostic I: Covariate Balance

- If people are sorting in the neighborhood of the threshold, we'd expect to see imbalance in distributions of observed covariates
- This motivates a check for whether there is a discontinuity in $E\left[X_i | R_i = r\right]$ at $R_i = c$
- Exactly analogous to a balance check in an RCT
- Implement by using X_i as the RD dependent variable

RD Diagnostic II: Bunching

- McCrary (2008) proposes a test for sorting that might compromise the validity of an RD design
- If individuals are strategically locating above or below the threshold, we'd expect "bunching" on whichever side of the discontinuity is preferable
- More generally, strategic manipulation may generate anomalies in the distribution of R_i around the threshold
- McCrary suggests looking for a discontinuity in the density of R_i near $R_i = c$
- This is often implemented visually
- Bunching around the threshold compromises RD estimates of treatment effects, but reflects a behavioral response that may be of substantive interest, see Saez's research on bunching at kink point of tax schedule for instance.

Urquiola and Verhoogen (2009)

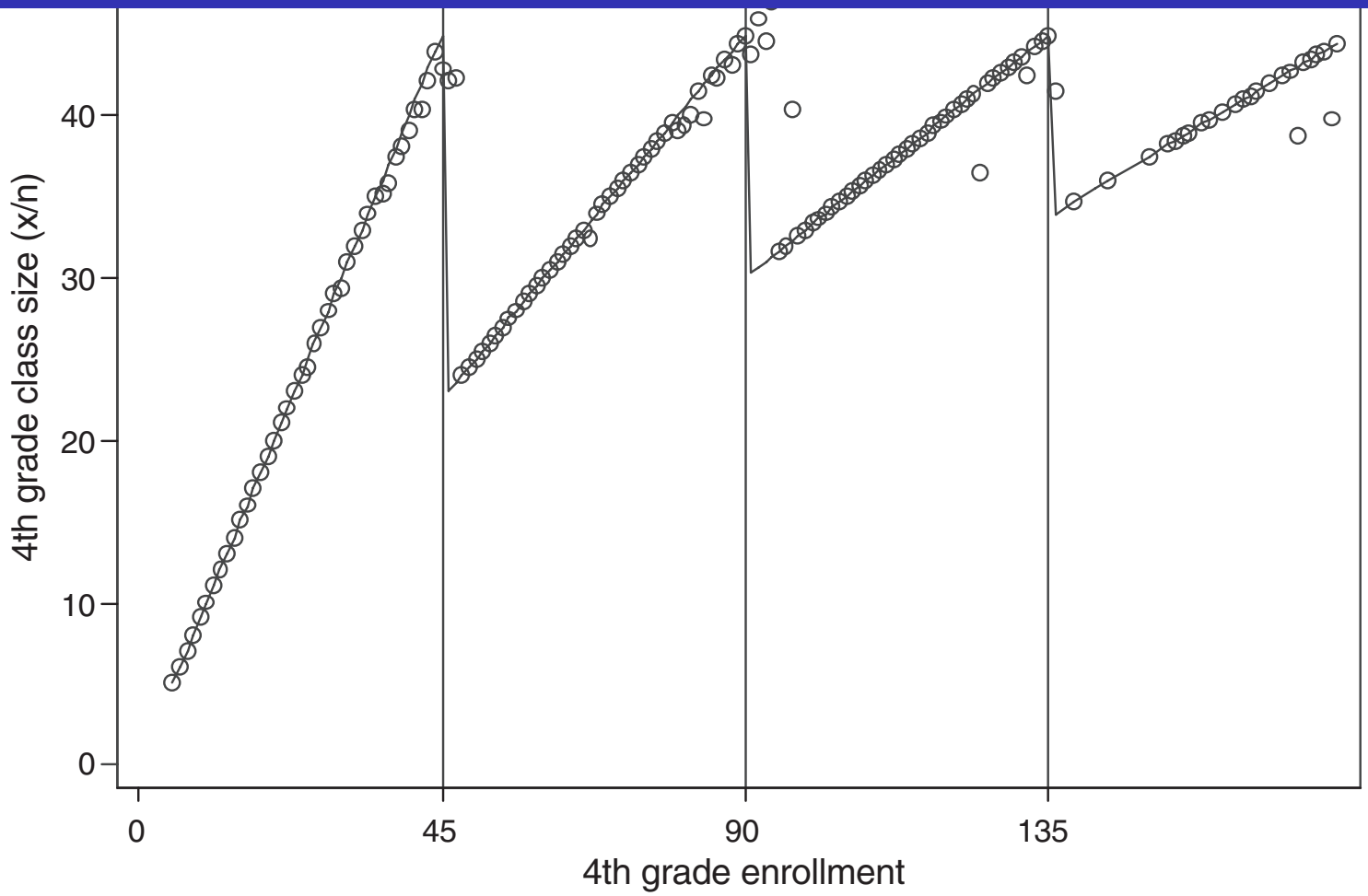
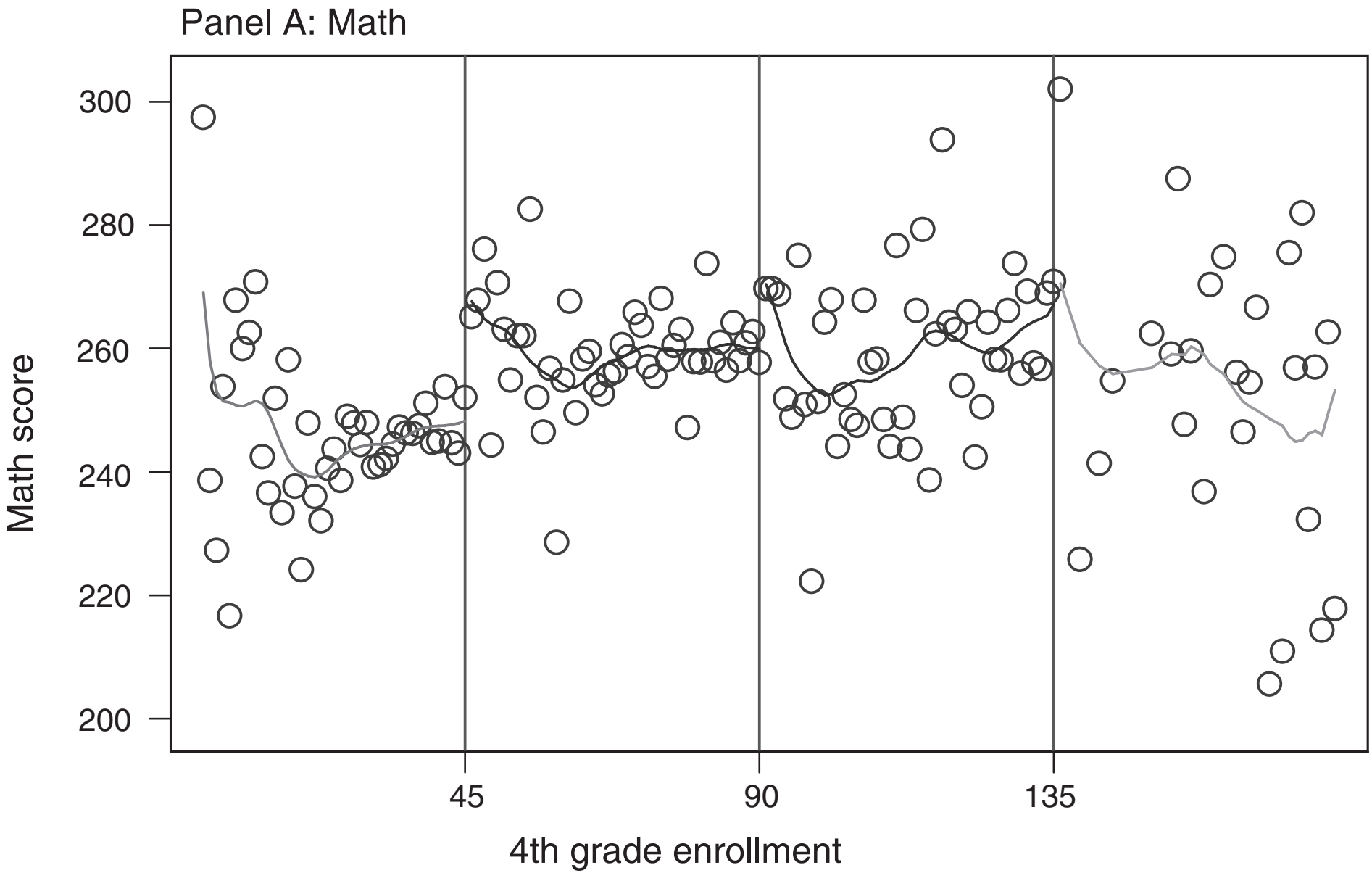
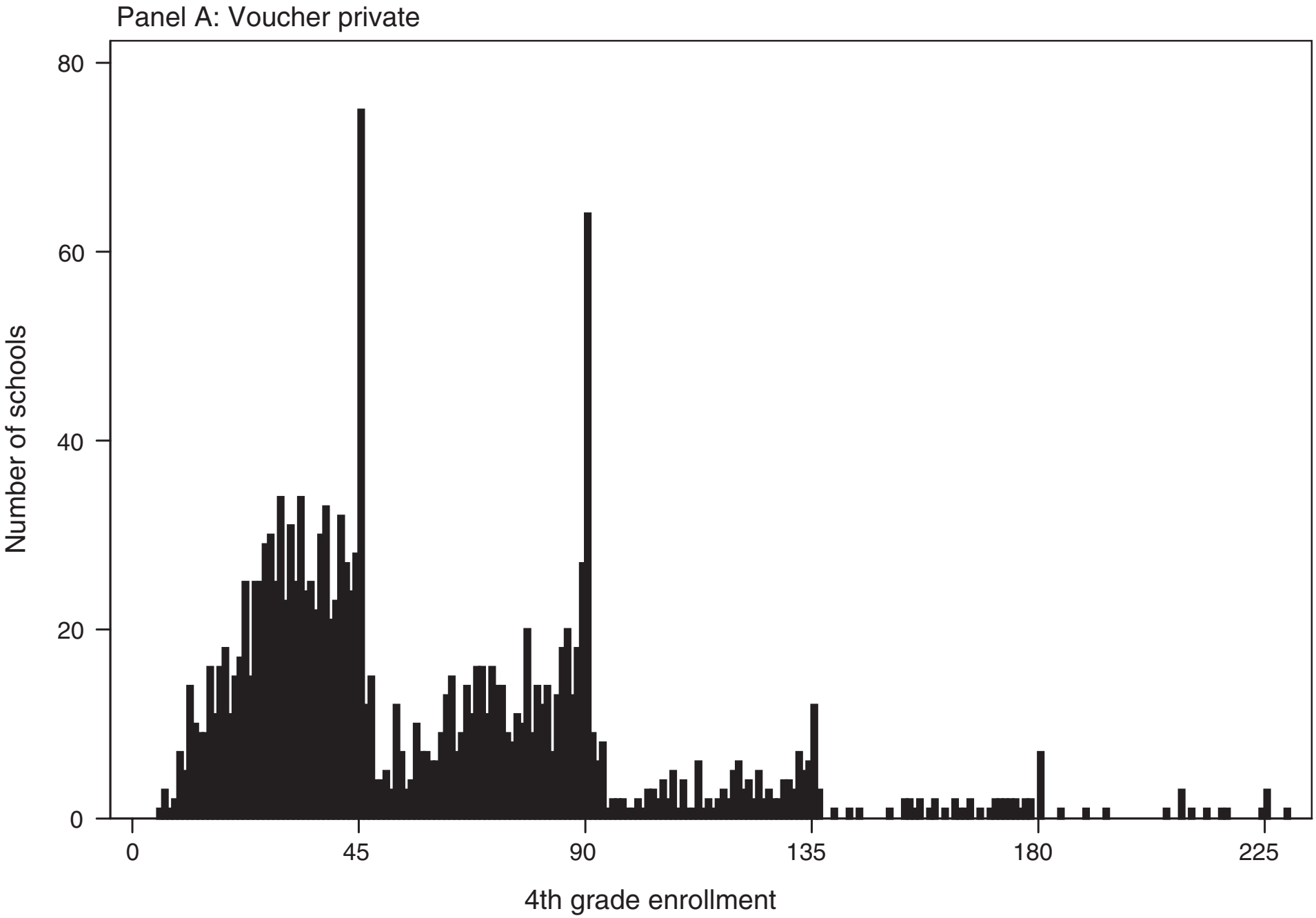
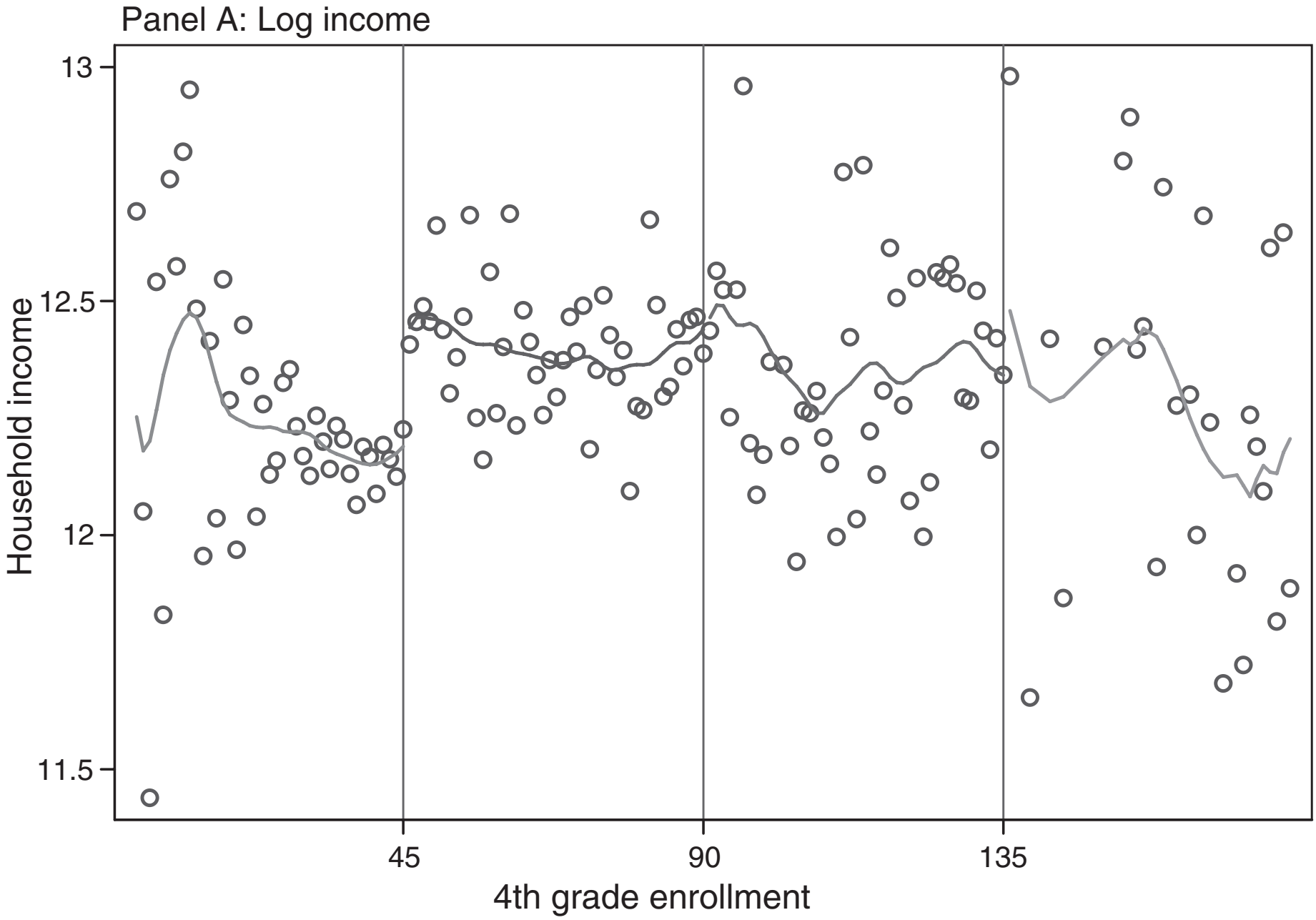


FIGURE 5. FOURTH GRADE ENROLLMENT AND CLASS SIZE IN URBAN PRIVATE VOUCHER SCHOOLS, 2002

Notes: Based on administrative data for 2002. The solid line describes the relationship between enrollment and class size that would exist if the class size rule (equation (30) in the text) were applied mechanically. The circles plot actual enrollment cell means of fourth grade class size. Only data for schools with fourth grade enrollments below 180 are plotted; this excludes less than 2 percent of all schools.


Panel A: Math
Math score
300
280
260
240
220
200
45
90
135
4th grade enrollment


Panel A: Voucher private
Number of schools
0
20
40
60
80
0
45
90
135
180
225
4th grade enrollment


Panel A: Log income
13
12.5
12
11.5
Household income
45
90
135
4th grade enrollment

RD Estimation: Global vs. Local

- Let's return to our discussion of RD implementation
- When should we use a global polynomial approach vs. a nonparametric local approach?
- There is no real conceptual distinction: we need to choose a bandwidth and a polynomial order
- Global estimators use an infinite bandwidth and a relatively high-order polynomial
- Local estimators use a smaller bandwidth and a lower-order polynomial (often linear)

RD Estimation: Global vs. Local

- Global estimators can rely heavily on data far from the discontinuity (Gelman and Imbens 2014, "Why high-order polynomials should not be used in regression discontinuity designs")
- If the outcome CEF is well-approximated by a polynomial, then using points far from the discontinuity to estimate the polynomial is legitimate and increases precision. If not, the global approach may perform poorly
- Not much is known about choosing the optimal order of a global polynomial for estimating the causal effect of interest; in contrast, there is a large literature on optimal bandwidths for local estimators of treatment effects
- Most researchers have gravitated towards the local approach

Kernels and Bandwidths

- Suppose we want to run local linear regression. What kernel and bandwidth should we choose?
- Many researchers use the "edge" kernel:

$$K(u) = 1\{|u| \leq 1\} \cdot (1 - |u|)$$

- The edge kernel has optimality properties in boundary estimation problems (Cheng et al., 1997)
- Also intuitively appealing: generates weighting function

$$K\left(\frac{R_i - c}{h}\right) = 1\{|R_i - c| \leq h\} \cdot \left(1 - \frac{|R_i - c|}{h}\right)$$

- The bandwidth h can then be interpreted as cutoff distance beyond which data are not used, and weights fall linearly from 1 to 0 in remaining sample

Optimal Bandwidth

- Recent literature focuses on optimal choice of bandwidth h
- Bias/variance tradeoff: Smaller bandwidth reduces bias from using points away from the boundary, but increases precision
- Intuitively, if there is not a lot of curvature in the CEF of Y_i, the bias from using points away from the boundary to estimate a regression will be small
- Imbens and Kalyanaraman (IK, 2012) use an asymptotic approximation to the mean squared error of the RD estimator and derive the MSE-minimizing bandwidth
- The optimal bandwidth depends on the curvature of the CEF near the discontinuity – IK propose to use plug-in estimators of parameters governing curvature

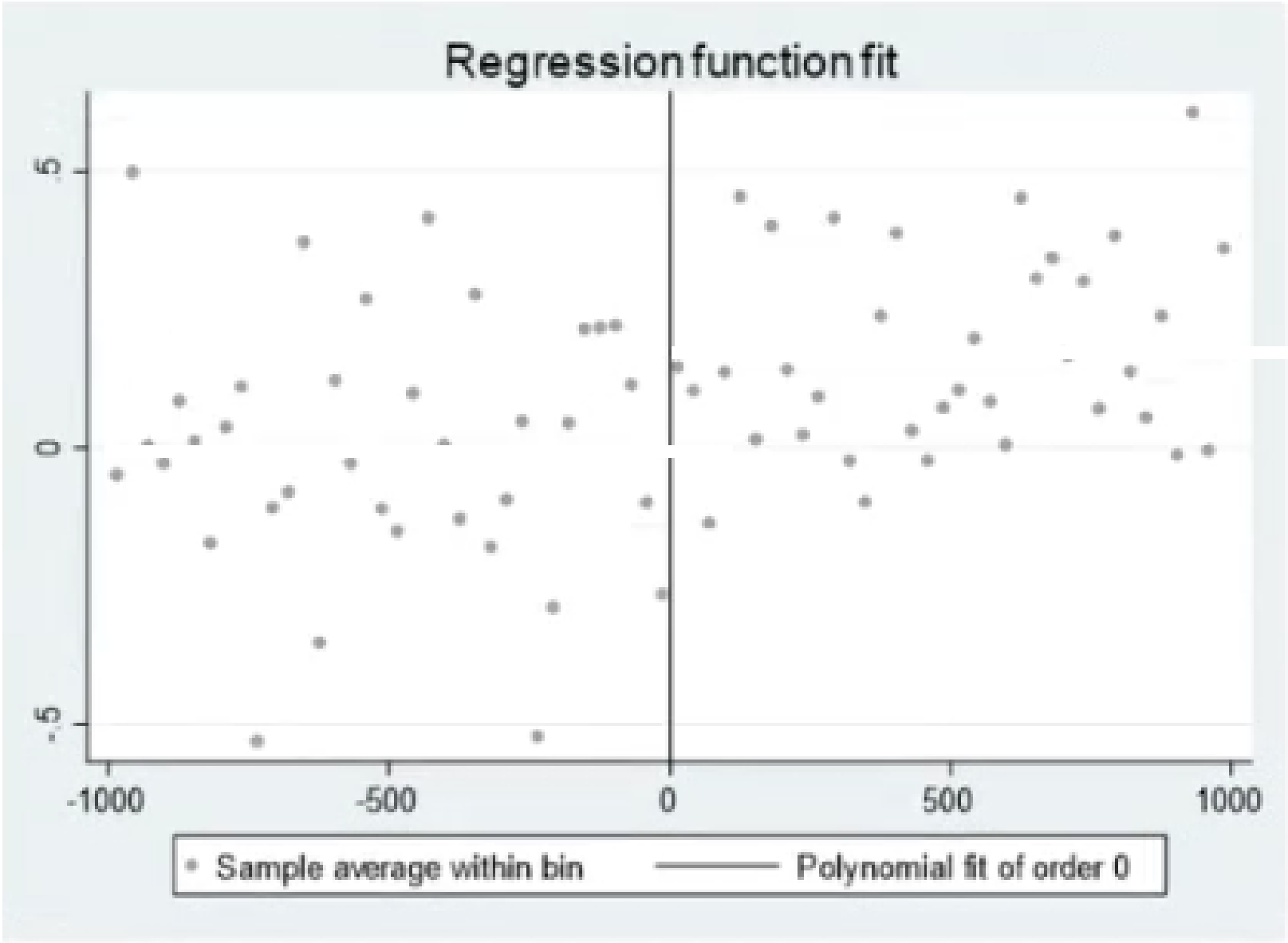
Robust Confidence Intervals

- The IK bandwidth minimizes MSE and is therefore well-suited to estimation
- Calonico, Cattaneo and Titiunik (CCT, 2014) show that it is poorly suited for inference, however: the IK bandwidth leaves an asymptotically non-negligible bias term in the estimate, so naive inference can lead to misleading confidence intervals
- CCT advocate using a second, smaller bandwidth that removes this bias term when constructing confidence intervals

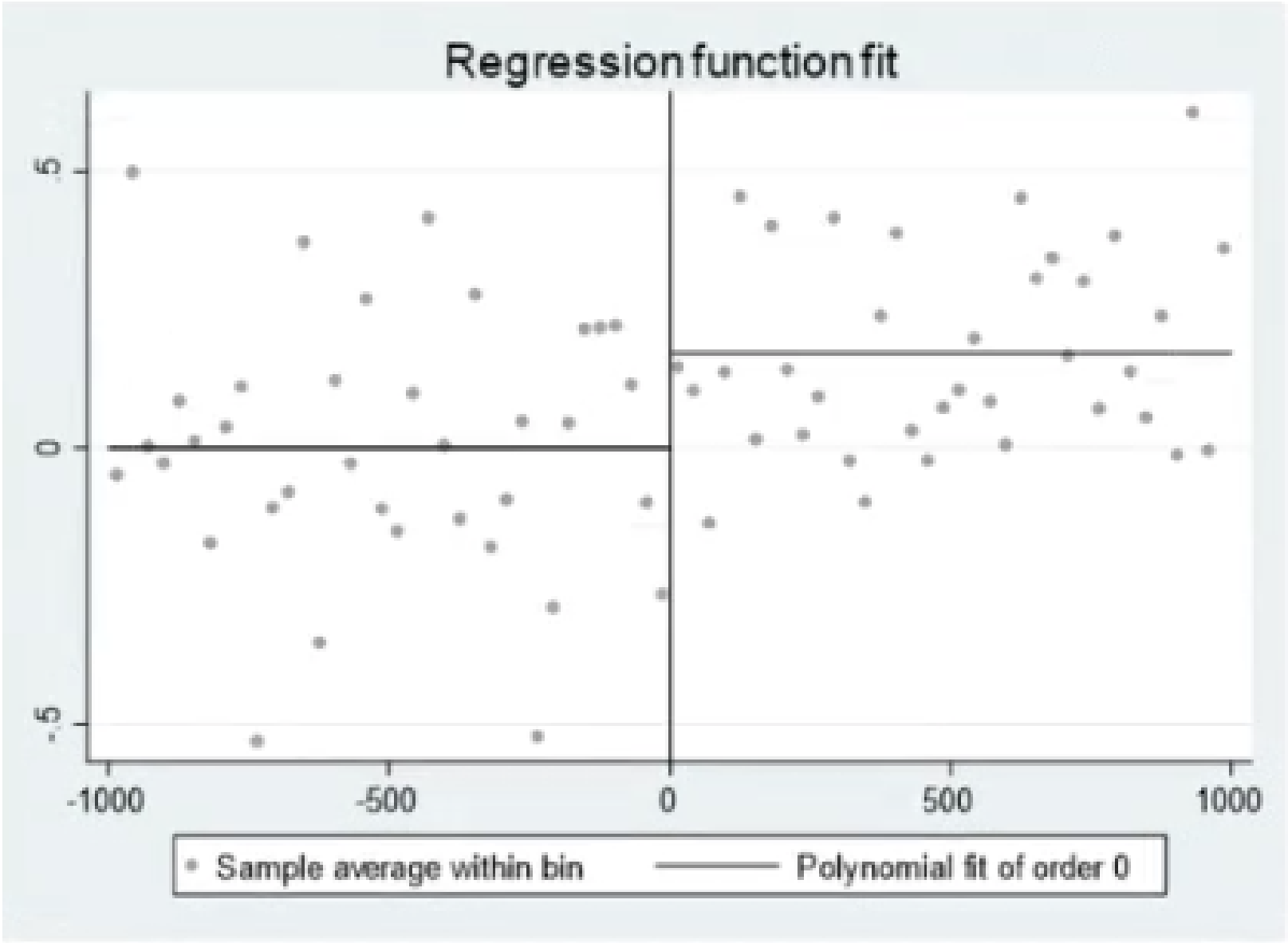
Robust Confidence Intervals

- Unfortunately, the CCT bandwidth is typically much smaller than the IK bandwidth and generates large confidence intervals
- The optimal bandwidth literature may limit the feasibility of RD projects in practice since lots of data is required to detect an effect with CCT's robust confidence intervals
- The IK bandwidth, CCT confidence intervals, and corresponding estimators for fuzzy RD designs are automated in the rdrobust stata package

Is this a compelling RD-plot?



Can you guys guess what is the t-stat in this graph?



RD with a Discrete Running Variable

RD with a Discrete Running Variable

D.S. Lee, D. Card / Journal of Econometrics 142 (2008) 655–674

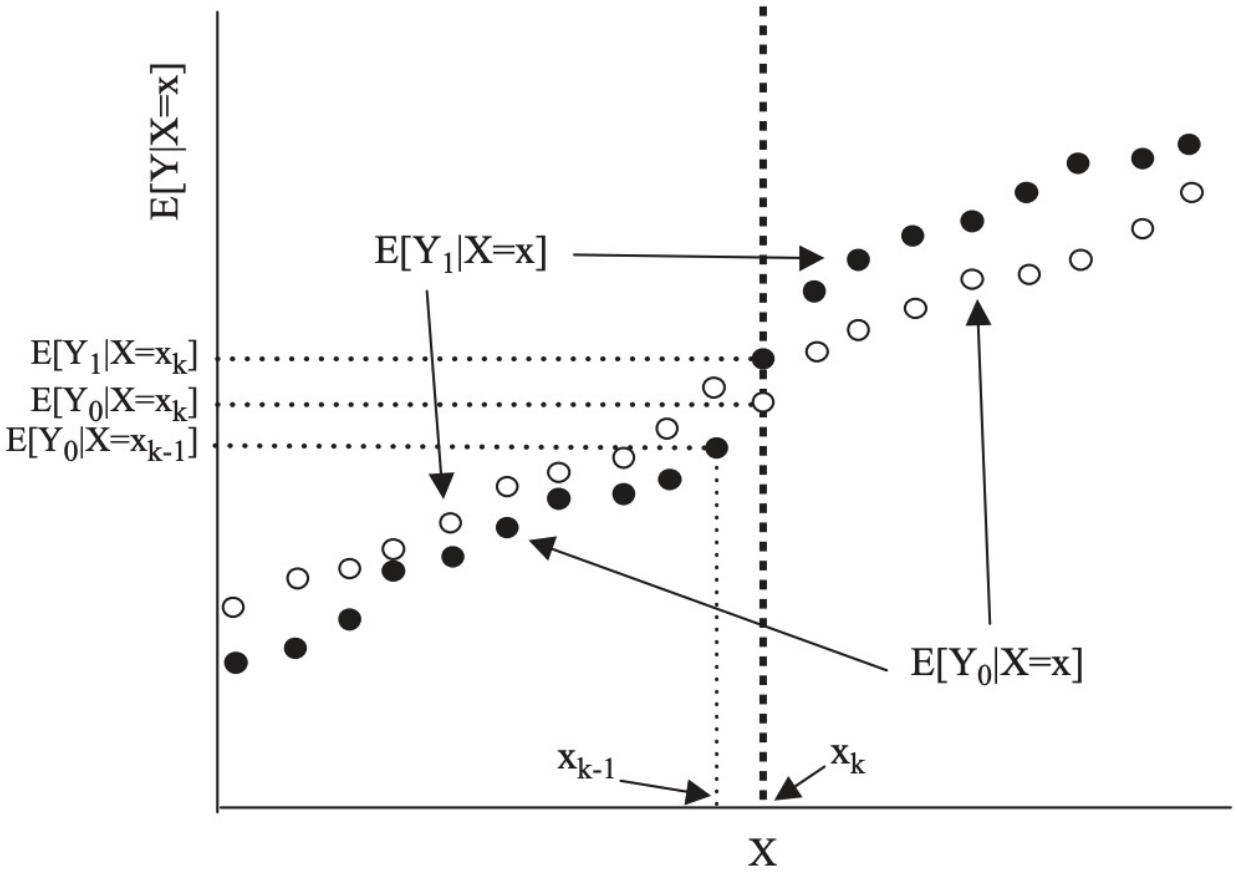


Fig. 2. Regression discontinuity, discrete covariate.

RD with a Discrete Running Variable

- In some cases the running variable is discrete rather than continuous
- This can occur either because the true running variable is continuous but the data are grouped, or because assignment occurs at the group level
- Example: Medicare eligibility begins at age 65, but age is typically recorded in months or years
- With data of this type it is not possible to compare outcomes in arbitrarily small neighborhoods of the cutoff; identification requires some form of parametric extrapolation
- Card and Lee (2008): recommend clustering on values of the running variable in settings with a group-level running variable

Sitting on the frontier

Inference in Regression Discontinuity Designs with a Discrete Running Variable†

By Michal Kolesár and Christoph Rothe*

We consider inference in regression discontinuity designs when the running variable only takes a moderate number of distinct values. In particular, we study the common practice of using confidence intervals (CIs) based on standard errors that are clustered by the running variable as a means to make inference robust to model misspecification (Lee and Card 2008). We derive theoretical results and present simulation and empirical evidence showing that these CIs do not guard against model misspecification, and that they have poor coverage properties. We therefore recommend against using these CIs in practice. We instead propose two alternative CIs with guaranteed coverage properties under easily interpretable restrictions on the conditional expectation function. (*JEL* C13, C51, J13, J31, J64, J65)

Sitting on the frontier

Simple and Honest Confidence Intervals in Nonparametric Regression*

Timothy B. Armstrong† Michal Kolesár‡

Yale University Princeton University

August 30, 2019

Abstract

We consider the problem of constructing honest confidence intervals (CIs) for a scalar parameter of interest, such as the regression discontinuity parameter, in nonparametric regression based on kernel or local polynomial estimators. To ensure that our CIs are honest, we use critical values that take into account the possible bias of the estimator upon which the CIs are based. We show that this approach leads to CIs that are more efficient than conventional CIs that achieve coverage by undersmoothing or subtracting an estimate of the bias. We give sharp efficiency bounds of using different kernels, and derive the optimal bandwidth for constructing honest CIs. We show that using the bandwidth that minimizes the maximum mean-squared error results in CIs that are nearly efficient and that in this case, the critical value depends only on the rate of convergence. For the common case in which the rate of convergence is $n^{-2/5}$, the appropriate critical value for 95% CIs is 2.18, rather than the usual 1.96 critical value. We illustrate our results in a Monte Carlo analysis and an empirical application.

RD Example: Clark and Martorell (2014)

- Clark and Martorell (JPE 2014) use an RD design to estimate the causal effect of high school graduation on earnings
- Two views on the causal effect of schooling on earnings:
  - Human capital: Schooling raises productivity
  - Signaling: Schooling reveals ability but has no productive value
- OLS returns to education are especially large for grade 12
- How much of this "sheepskin effect" reflects signaling?

Clark and Martorell (2014)

- CM use the fact that students in Texas must pass exams before graduating high school
- Testing starts in 10th grade and students can try multiple times, but eventually face a "last chance" exam at the end of 12th grade
- Students who just barely fail vs. barely pass should have similar human capital, but differ in educational credentials
- RD therefore plausibly identifies the signaling value of a diploma
- There is some "slippage" even with last-chance exams – so the RD is fuzzy

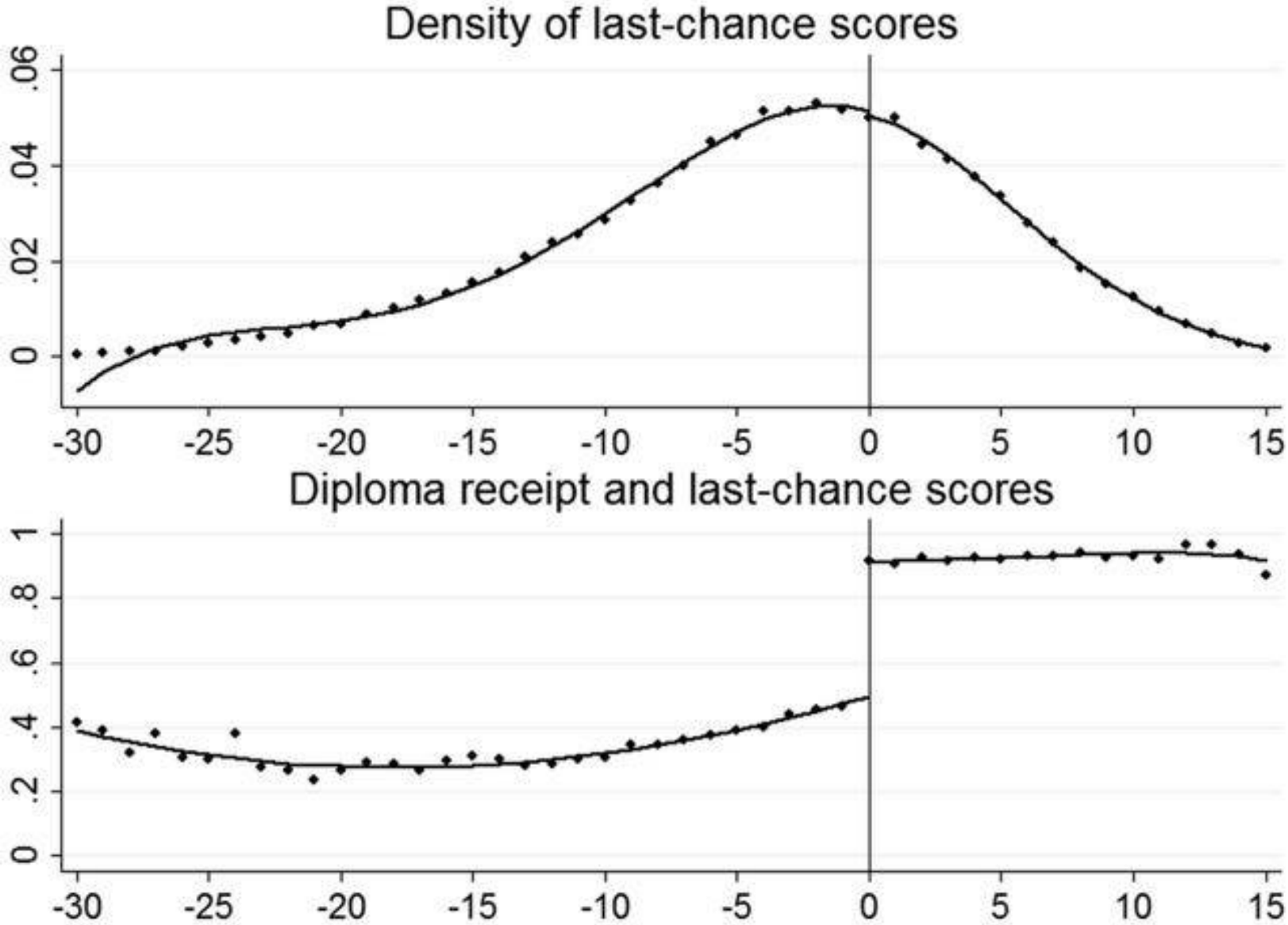


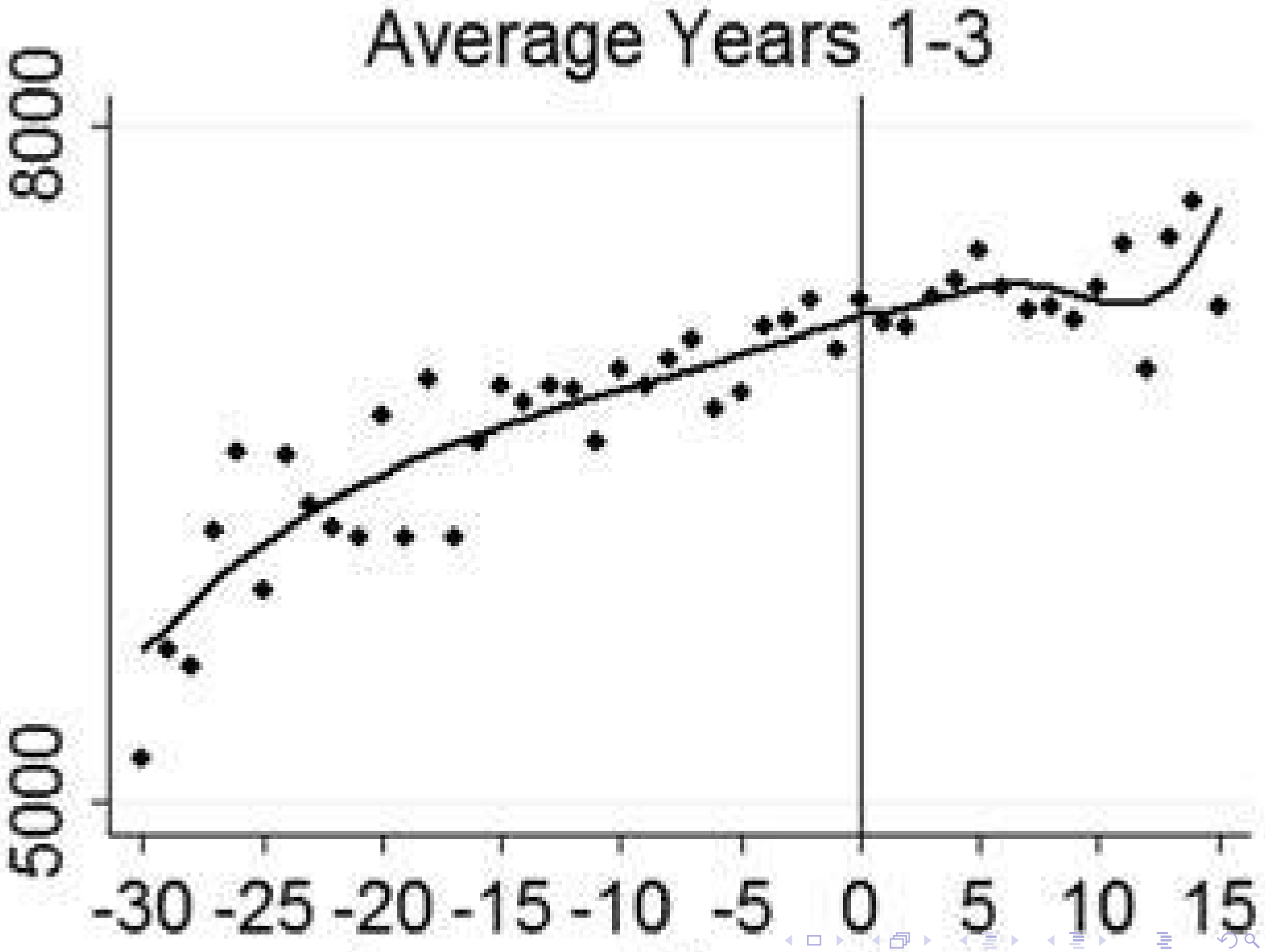
FIG. 1.—Last-chance exam scores and diploma receipt. The graphs are based on the last-chance sample. See table 1 and the text. Dots are test score cell means. The scores on the *x*-axis are the minimum of the section scores (recentered to be zero at the passing cutoff) that are taken in the last-chance exam. Lines are fourth-order polynomials fitted separately on either side of the passing threshold.

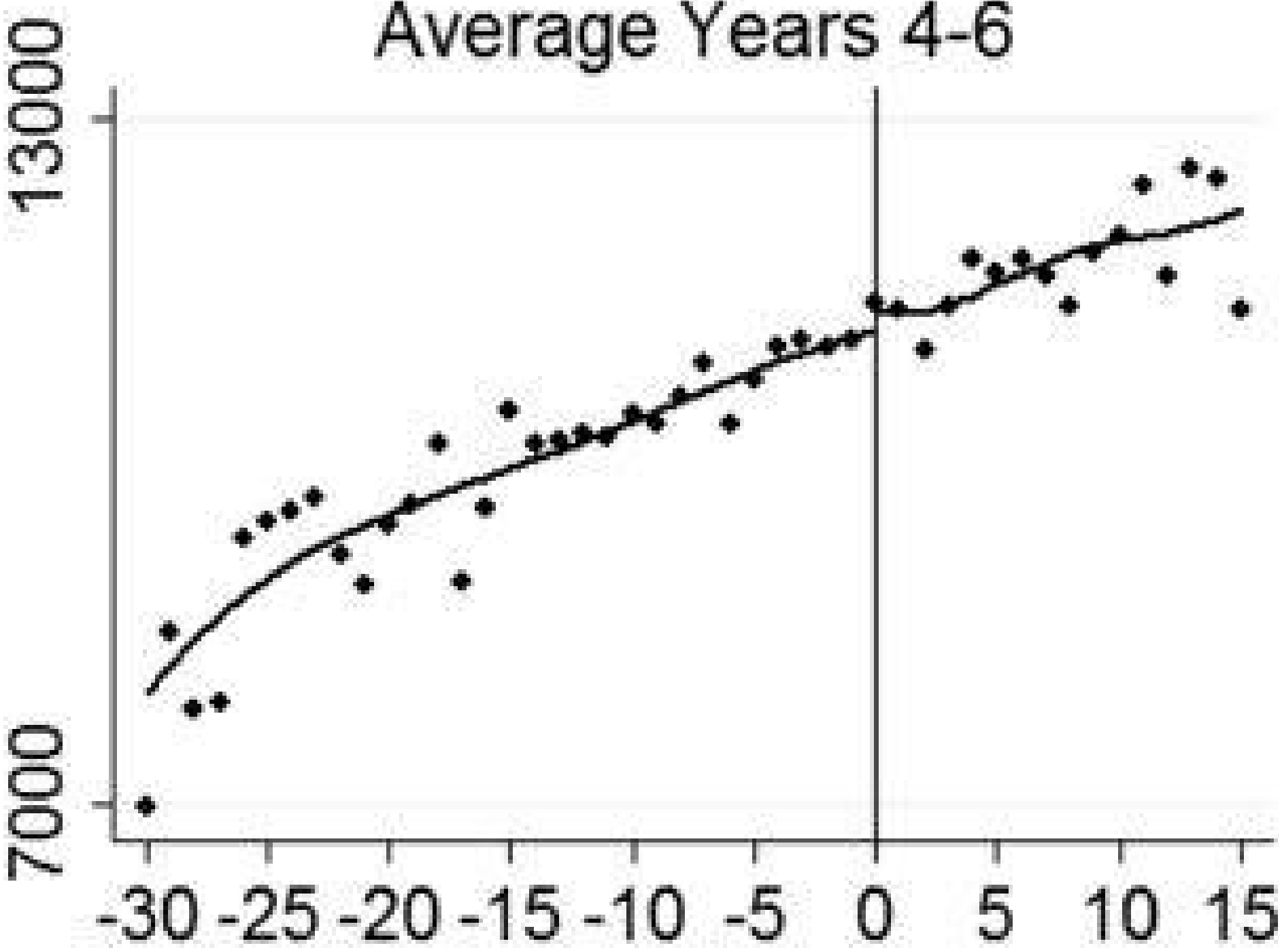
TABLE 2

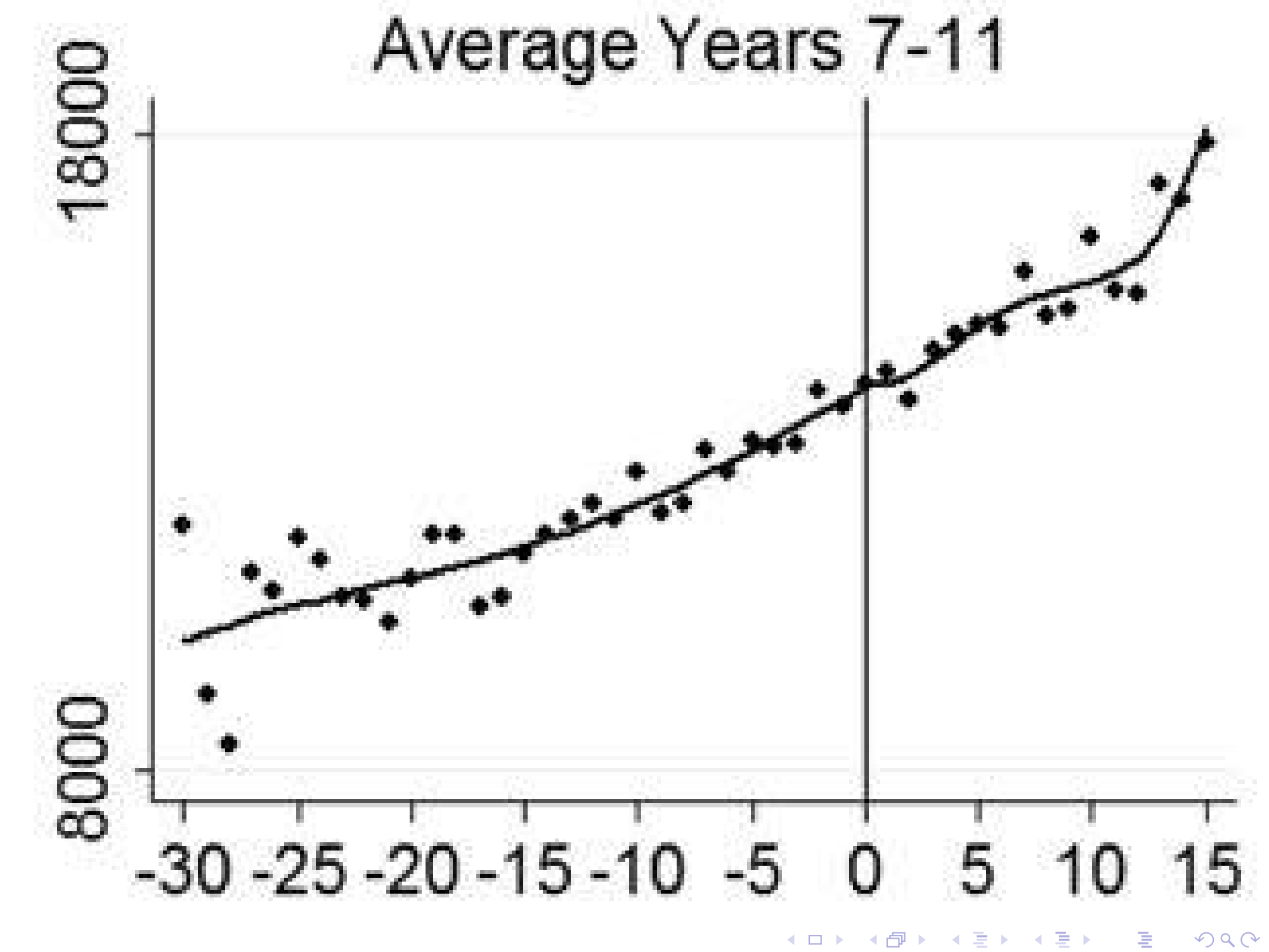
IMPACT OF PASSING THE LAST-CHANCE EXAM ON THE PROBABILITY OF EARNING A DIPLOMA

Receive High School Diploma	(1)	(2)	(3)	(4)	(5)
By end of summer after 12th grade (sample mean = .363)	.545	.484	.481	.475	.486
	(.007)	(.009)	(.012)	(.016)	(.009)
Within 1 year of last-chance exam (sample mean = .452)	.480	.420	.425	.424	.422
	(.007)	(.009)	(.012)	(.016)	(.009)
Within 2 years of last-chance exam (sample mean = .465)	.472	.415	.419	.417	.417
	(.007)	(.009)	(.012)	(.016)	(.009)
Within 3 years of last-chance exam (sample mean = .468)	.468	.412	.416	.414	.414
	(.007)	(.009)	(.012)	(.016)	(.009)
Baseline covariates?	No	No	No	No	Yes
Degree of test score polynomial	1	2	3	4	2

NOTE.—The table is based on last-chance samples (see table 1 and the text). "Degree of test score polynomial" refers to the test score polynomials controlled for in these regressions (all interacted with a dummy for passing the exam). Column 5 presents estimates based on models that also control for covariates (see note to table 1). Robust standard errors are in parentheses. There are 37,571 observations in each panel.


Average Years 1-3
8000
5000
-30
-25
-20
-15
-10
-5
0
5
10
15


Average Years 4-6
13000
7000
-30
-25
-20
-15
-10
-5
0
5
10
15


Average Years 7-11
18000
8000
-30
-25
-20
-15
-10
-5
0
5
10
15

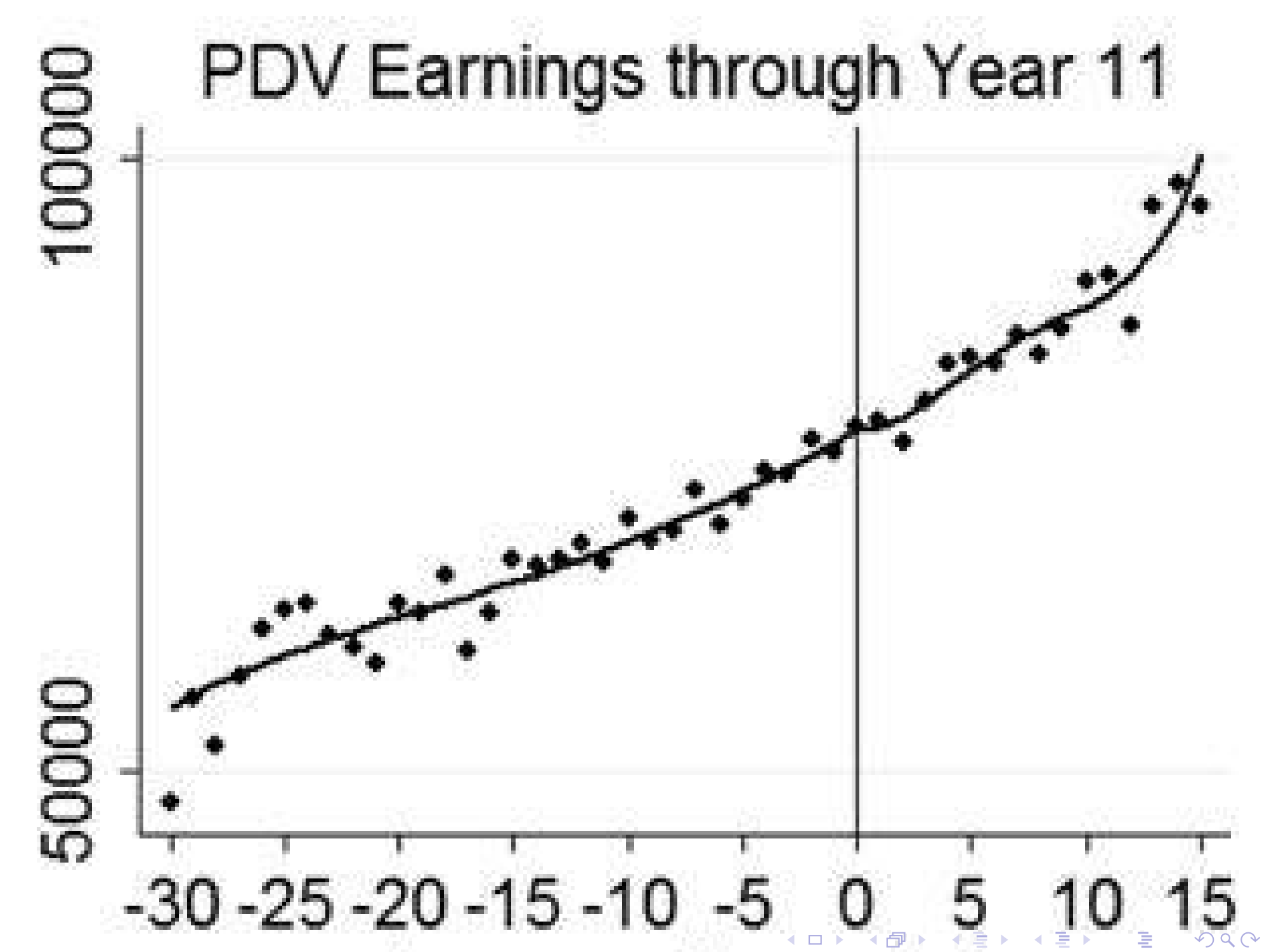

PDV Earnings through Year 11
100000
50000
-30
-25
-20
-15
-10
-5
0
5
10
15

TABLE 4

IMPACT OF RECEIVING A HIGH SCHOOL DIPLOMA ON EARNINGS

	(1)	(2)
Years 1–3	109.5	119.8
	(352.0)	(283.2)
	1.6%	1.7%
Years 4–6	333.1	155.0
	(517.9)	(421.7)
	3.0%	1.4%
Years 7–11	99.3	52.2
	(767.0)	(634.0)
	.7%	.4%
All years pooled	177.7	106.4
	(475.9)	(392.9)
	1.7%	1.0%
PDV earnings through year 11	1,632.1	598.8
	(3,299.5)	(2,742.3)
	2.1%	.8%
Polynomial specification	Unrestricted	Restricted

NOTE.—Estimates use the same samples and specification as cols. 5 and 6 of table 3. Unrestricted specification estimates are from two-stage least-squares models in which the test score polynomial is fully interacted with an indicator for passing the last-chance exam. Results from the "restricted" polynomial specification are the ratio of the reduced-form and first-stage estimates, where these are estimated separately and the standard errors are calculated using the delta method. The reduced form is the estimated discontinuity in earnings using the polynomial specification in which the slopes are constrained to be equal on either side of the passing cutoff (col. 6 in table 3) and the first stage is estimated using a polynomial that is fully interacted with the passing dummy. All standard errrors are adjusted for clustering at the individual level. For each set of estimates, the third row represents the point estimate expressed as a percentage of mean earnings just to the left of the passing cutoff.

TABLE 5

SIGNALING VALUES OF A HIGH SCHOOL DIPLOMA FOR SUBGROUPS

	By Gender		By Race		By County: Mobility and Federal Employment	
	Male	Female	White	Nonwhite	Low	High
Years 1–3	188.1	91.1	−8.9	154.1	137.9	89.7
	(546.8)	(303.6)	(561.1)	(322.6)	(393.7)	(407.4)
Years 4–6	452.0	8.7	−228.2	247.0	556.0	−276.6
	(814.2)	(449.9)	(838.7)	(480.9)	(584.9)	(607.5)
Years 7–11	492.0	−138.6	−124.4	19.8	−149.7	329.1
	(1,224.5)	(655.6)	(1,269.6)	(703.3)	(878.2)	(916.1)
Years pooled	383.5	−17.7	−120.7	135.8	163.7	57.3
	(765.5)	(407.2)	(791.3)	(438.5)	(551.6)	(557.7)
PDV earnings through year 11	2,473.8	−297.0	938.5	−1,457.0	1,037.9	114.7
	(5,391.4)	(2,824.2)	(3,035.7)	(5,621.4)	(3,885.9)	(3,849.0)

NOTE.—Estimates are from the preferred restricted polynomial specification with baseline covariates. See the text and note to table 4 for additional details.

Encore

Multiple Discontinuities

- Often more than one discontinuity present.
- Nonparametric solution is to estimate each RD separately then average across them as desired. (see Black, Galdo, Smith 2007 AER)
  - Only feasible if discontinuities far enough apart to get a good estimate on either side
- Second approach is to impose some sort of stationarity – e.g. running variable becomes distance to nearest discontinuity.
- Third approach is to go fully parametric.

See Angrist and Lavy (1999) for an example.

Regression Kink Design

- Recent extension of RD: The **regression kink design** (RKD; Card et al., 2015)
- Instead of exploiting a discontinuity in the CEF of the treatment variable, the regression kink design exploits a kink in the CEF of a continuous treatment (i.e. a discontinuity in the first derivative)
- A corresponding kink in the distribution of the outcome variable suggests the presence of a treatment effect

Regression Kink Design

- Suppose the treatment of interest is a deterministic function of the running variable:

$$S_i = b(R_i)$$

- Here $b(\cdot)$ is a continuous function with a kink at c
- Let $f_i(s)$ denote i's potential outcome as a function of the treatment. The observed outcome is

$$Y_i = f_i(S_i)$$

Regression Kink Design

- Then under mild regularity conditions:

$$\frac{\lim_{r \to c^+} \frac{dE[Y_i|R_i=r]}{dr} - \lim_{r \to c^-} \frac{dE[Y_i|R_i=r]}{dr}}{\lim_{r \to c^+} b'(r) - \lim_{r \to c^-} b'(r)} = E\left[f_i'(S_i)|R_i = c\right]$$

- The ratio of the discontinuity in the outcome derivative to the discontinuity in the treatment derivative identifies the average marginal effect of treatment for individuals at the threshold
- As before, the key assumption is that potential outcomes are smooth around the threshold – any kink in the outcome CEF must be due to the treatment
- Diagnostics: Look for kinks in covariate distributions, or bunching in the density of R_i

Fuzzy Regression Kink Design

- As with RD, we can generalize RKD to a "fuzzy" scenario where the treatment is not a deterministic function of R_i

$$S_i = b_i(R_i)$$

- $b_i(r)$ may not depend on r for some individuals, and may have a continuous derivative for others
- Monotonicity condition:

$$\lim_{r \to c^+} b_i'(r) \geq \lim_{r \to c^-} b_i'(r) \ \forall i$$

- All kinks in the treatment assignment rule work in the same direction

Fuzzy Regression Kink Design

- Then under appropriate regularity conditions:

$$\frac{\lim_{r \to c^+} \frac{dE[Y_i|R_i=r]}{dr} - \lim_{r \to c^-} \frac{dE[Y_i|R_i=r]}{dr}}{\lim_{r \to c^+} \frac{dE[S_i|R_i=r]}{dr} - \lim_{r \to c^-} \frac{dE[S_i|R_i=r]}{dr}} = E\left[\omega_i \cdot f_i'(S_i)|R_i = c\right]$$

- ω_i can be thought of as a "complier" weight that captures the extent to which i responds to the kink
- Estimation issues with RKD parallel those with RD
- Can be implemented via local polynomial regression with the analogue of the IK bandwidth and CCT robust CI
- Data requirements for nonparametric RKD estimation are substantial

RKD Estimation

- Estimation issues with RKD parallel those with RD
  - Can be implemented via local polynomial regression with the analogues of the IK bandwidth and CCT robust CI
  - rdrobust provides an implementation
- Corresponding diagnostics
  - Visualization
  - Kinks in baseline covariates?
- Need even more data to precisely estimate kinks than to estimate discontinuities